

Brayan Elian Castiblanco

Aarhus, Denmark • bcastiblanco@post.au.dk • (+45) 31848024

EDUCATION

Erasmus Mundus Master in Quantum Technologies and Engineering (QuanTEEM) Sep 2024 – Aug 2026

- **MSc Physics** — Aarhus University, Denmark Aug 2025 – Jun 2026
GPA: 9.1 MSc Thesis Grade: 10/12 (7-point grading scale: 12 highest, -3 lowest)
- **MSc Advanced Quantum Physics** — RPTU Kaiserslautern, Germany Feb – Jul 2025
GPA: 2.4 (1.0 highest, 6.0 lowest)
- **Master in Fundamental Physics and Appl.** — Université Bourgogne Europe, France Sep 2024 – Jan 2025
GPA: 16.24 (20.00 highest, 0.00 lowest)

MSC THESIS: “Ultrafast dynamics and non-linear light-induced phase transitions in Tellurium”.

BSc Physics — National University of Colombia Feb 2017 – Jun 2022

GPA: 4.2 (5.0 highest, 0.0 lowest) | Meritorious BSc thesis

BSC THESIS: “Thermal leptogenesis in the type-I Dirac extension to the DFSZ axion model for dark matter”.

COURSEWORK: Platforms for Quantum Technologies, Quantum Engineering and Information, Photonic Integrated Circuits, Advanced Solid-State Physics, Quantum Optics I & II, Atomic and Molecular Physics.

RESEARCH EXPERIENCE

“Ultrafast Dynamics of Quantum Materials” Group — Dept. of Physics and Astronomy, Aarhus University

Associate Research Student Jan – Jun 2026

- Carried out table-top pump-probe and rotational anisotropy spectroscopy experiments in the near-infrared spectrum, under UHV and cryogenic conditions.
- Extracted coherent phonon frequencies, amplitudes, and damping rates from linear and nonlinear optical reflectance data by fitting time-domain signals to Raman- and IR-active phonons in a tensor model.
- Skilled in non-linear optics phenomena: Second Harmonic Generation (SHG), Sum and Difference Frequency Generation (SFG/DFG), in crystals in free-space optics experimental setups.
- Applied group symmetry analysis to the dielectric function and non-linear susceptibility tensor in crystals with arbitrary orientation, characterising their phonon spectra and optical anisotropy response.

SUPERVISOR: Prof. Dr. Simon Wall (simon.wall@phys.au.dk)

Center for Complex Quantum Systems — Dept. of Physics and Astronomy, Aarhus University

Research Intern Aug 2025 – Feb 2026

- Analyzed linear spin-wave theory for Heisenberg antiferromagnets, in the nearest and next-nearest neighbor models, deriving analytic dispersion and magnetization expressions for arbitrary geometries.
- Analyzed phase transitions on Heisenberg lattices from frustration effects.
- Built Python tools for magnon dispersion and regularized Brillouin-zone integration, benchmarked staggered magnetization against quantum Monte Carlo and series expansion data. (See repository [here](#))

SUPERVISOR: Prof. Dr. Georg Bruun (bruungmb@phys.au.dk)

“Theoretical Quantum Optics Group” — Physics Department, RPTU Kaiserslautern

Research Intern Feb – Jul 2025

- Simulated band structures and energy evolution in topological quantum transport systems: Thouless pumping, Aubry-André-Harper model, and non-linear Kerr solitons.
- Developed a theoretical model reproducing integer transport in 1D arrays of coupled waveguides.
- Calculated Zak phase and Chern numbers to characterize soliton solutions; developed numerical tools for soliton dynamics and topological properties in optical lattice models. (See repository [here](#))

SUPERVISOR: Prof. Dr. Michael Fleischhauer (mfleisch@physik.uni-kl.de)

“Field and Particles” Research Group — Dept. of Physics, National University of Colombia

Research Assistant (Part Time) Sep 2021 – Aug 2023

- Investigated QCD axions as dark matter candidates within thermal leptogenesis models via Peccei–Quinn symmetry breaking in an accelerating universe model. (See BSc thesis [here](#))
- Analyzed cosmological relic abundance dependence on baryon density, neutrino, and axion masses.

“Experimental Particle Physics” Internship Project — Dept. of Physics, National University of Colombia

Project Author

Oct 2021 – Feb 2022

- Reconstructed Higgs boson decay into Z-boson pairs from the 13 TeV ATLAS Open Data release; developed analysis notebooks in Python using the ROOT Data Analysis Framework. (See repository [here](#))

SUPERVISOR: Prof. Dr. Carlos Sandoval Usme (cesandovalu@unal.edu.co)

SKILLS

Computer:

- Advanced: Python, MATLAB, Linux, $\text{\LaTeX}$.
- Intermediate: Mathematica, EMode Photonix.
- Beginner: Qiskit, Maple, Tidy3D.

Languages:

- Spanish (native).
- English (C1 CEFR).
- German (A2 CEFR).

TEACHING, MENTORING, AND LEADERSHIP EXPERIENCE

School of Engineering — ECCI University, Bogota

Auxiliary Professor (Part time)

Jan – Aug 2024

- Delivered undergraduate courses in Fundamentals of Mathematics and Probability & Stochastic Processes.

Academic Department and University Welfare — Bogota District University Francisco Jose de Caldas

Project Coordinator and Teaching Assistant

Feb 2023 – Aug 2024

- Coordinated a 12-member interdisciplinary tutoring team; assisted around 15 students/class in Electromagnetism, Oscillations and Waves, and Modern Physics.

Academic Division — School of Science, National University of Colombia

Teaching Assistant

Mar 2021 – Dec 2022

- Tutored Special Mathematics for Physics, Electrodynamics, Calculus I, and Fundamentals of Mathematics.

ACADEMIC EXPERIENCES

International Schools

- **Quantum Information Science with Light and Matter Summer School 2026** (*poster session*)
Organized by: DTU Physics & DigiQ, in Helsingør, Denmark. Aug 17 – 21, 2026
- **DigiQ Paris Summer School 2026: Quantum Technologies and Simulations** (*website event*)
Organized by: École Normale Supérieure & Sorbonne University, in Paris, France. Jun 29 – Jul 2, 2026
- **Giambiagi Winter School on Cosmology** (*poster session*)
Organized by: ICTP & DF-UBA, in Buenos Aires, Argentina. Jul 17 – 28, 2023
- **2022 Perimeter Institute–SAIFR Journeys into Theoretical Physics** (*certificate*)
Organized by: Perimeter Institute & IFT-UNESP/ICTP–SAIFR, in São Paulo, Brazil. Aug 1 – 7, 2022

Workshops and Conferences

- **Impurity Problems in Quantum Mixtures: from Ultracold Gases to Electron Matter** (*poster session*)
Organized by: ICTP, in Trieste, Italy. Feb 23 – 25, 2026
- **Quantum Technology Hackathon 2026: Exploring the Future of Quantum Technologies** (*repository*)
Organized by: DTU Physics, in Copenhagen, Denmark. Jan 26 – 30, 2026
- **Quantum Campus Aarhus Annual Meeting 2025** (*poster session, No. 12*)
Organized by: Aarhus University, in Aarhus, Denmark. Dec 9, 2025
- **CoCo 2022 Workshop: Cosmology in Colombia** (*speaker*)
Organized by: National Observatory of Colombia & CoNHEP, in Bogotá, Colombia. Sep 14 – 15, 2022

AWARDS, ACADEMIC RECOGNITIONS AND MEMBERSHIPS

- Member of the “**DigiQ (EU)-Network**”, part of the team “Advocates for Quantum Technologies” (2025).
- Erasmus Mundus Master “**QuanTEEM**” scholarship (2024).
- **Meritorious bachelor’s thesis** at National University of Colombia (2022).
- “**Andrés Bello Distinction**” and scholarship. Top 0.02% (Colombia) in university admission exam (2016).